TalentAgent™ — Recruiter Platform

Execution & Implementation Plan

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Document Purpose

This document is the engineering and product execution plan for the **TalentAgent™** — the recruiter/HR side of the HireOrGetHired platform. It is structured for development scoping, vendor selection, cost modelling, and sprint planning. Each module defines the feature in detail, the AI agents required (with their tasks, tools, and LLM models), the integrations needed, the technical components to build, and cost and complexity drivers.

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1. Platform Overview & Architecture Context

What TalentAgent™ Is

TalentAgent™ is the recruiter's persistent AI orchestration layer. It sits above their existing ATS (Greenhouse, Lever, Workday), connects to it via bidirectional API sync, and enriches every step of the hiring pipeline with agentic intelligence. It is not an ATS replacement — it is the intelligence and automation layer that makes the ATS exponentially more useful.

Core Architectural Decisions (Relevant to Recruiter Build)

Concern	Decision	Implication for Build
Frontend	React 18 + TypeScript + Tailwind CSS + Radix UI	Custom design system; accessible component library
State Management	Zustand	Lightweight; handles agent state streams and pipeline updates
Real-time	WebSocket (AWS API Gateway)	Live agent progress, pipeline changes, NegotiationCoach™ prompts
Charting	Recharts (pipeline analytics) + D3.js (Talent Intelligence Graph)	Two separate charting libraries required

Concern	Decision	Implication for Build
Auth	AWS Cognito + SAML 2.0/OIDC (Okta, Azure AD, Google Workspace)	Enterprise SSO is mandatory for recruiter product
Agent Orchestration	LangGraph (Python + FastAPI)	All agents are LangGraph state machines
ATS Integration	Bidirectional sync via connector microservices	ATS is always the system of record for HR compliance data
Multi-tenancy	Row-level security (PostgreSQL RLS) + per-tenant Pinecone namespaces	Strict enterprise data isolation
Compliance	Immutable audit log (PostgreSQL append-only) + database-level offer constraints	Cannot be bypassed at application layer
Human Handoff	Blocking async checkpoints in LangGraph — infrastructure-enforced	No agent action bypasses human approval at defined gates

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2. Module 1 — Requisition Intelligence

2.1 Feature Description

Recruiter describes a role conversationally in natural language. The system generates a structured job description, extracts skill requirements mapped to ESCO/O*NET taxonomy, benchmarks the compensation range against the Talent Intelligence Graph, and scores the requisition's "health" — flagging unrealistic requirements, seniority/comp mismatches, and suggesting adjustments. The requisition routes through a configured human approval chain before sourcing begins.

User-facing outputs:

- Structured JD generated from conversational description (editable)
- Required skills listed with ESCO/O*NET taxonomy IDs
- Comp benchmark widget: p25/median/p75/p90 for this role in this geography
- Requisition health score: green/amber/red with specific issue callouts
- Approval workflow: route to hiring manager → HR Ops → approved/rejected
- Approved requisition synced back to ATS

2.2 AI Agents Required

Agent 1.1 — `RequisitionIntelligenceAgent`

Property	Detail
Role	Generates structured JD from natural language input; benchmarks comp; scores requisition health
Trigger	Recruiter initiates new requisition via conversational input in dashboard
LLM Model	GPT-4o (high-quality JD generation; nuanced requirement extraction)
Orchestration	LangGraph conversational state machine; pauses at approval gate

Tasks:

JD Generation:

1. Receive natural language role description from recruiter (conversational input)
2. Extract structured fields: role title, seniority level, team, location, remote policy, key responsibilities, required skills, preferred skills, years of experience, education requirements
3. Map extracted skills to ESCO/O*NET canonical IDs using `skills_taxonomy_lookup_tool`
4. Generate formatted, well-structured JD in the recruiter's configured tone and style (enterprise customers can configure house style via prompt customisation)
5. Return to recruiter for review and edit before proceeding

Comp Benchmarking:

1. Query Neo4j Talent Intelligence Graph: `BENCHMARK (Role → MarketSignal)` for this role + geography + seniority (p25/median/p75/p90)
2. Query Levels.fyi data if tech role: total comp breakdown
3. Query Bureau of Labor Statistics API: occupational wage statistics
4. Cross-reference with company's own comp band data (from Workday HCM integration if available)
5. Display benchmark inline in JD editor; flag if proposed comp is below market median

Requisition Health Scoring:

1. Analyse skill requirements list: flag if requirements are contradictory, unrealistic for stated seniority, or commonly listed as "nice to have" for this role type
2. Check supply signal: query Neo4j for skill scarcity index for required skills in this geography — flag if combination is rare
3. Check comp vs. seniority: if comp is below market p25 for stated seniority level, flag as health risk
4. Generate health report: overall score (0–100), specific issues with suggested fixes
5. Return actionable suggestions: e.g., "Requiring 10+ years in Rust for a mid-level role will reduce your candidate pool by 94%. Suggest: 3+ years with demonstrated expertise."

Approval Workflow:

1. On recruiter approval of draft JD: create `PENDING_HUMAN_APPROVAL` checkpoint in LangGraph
2. Route approval notification to hiring manager (email via SES + in-app notification)
3. Hiring manager approves/rejects/requests changes via HOGH or linked ATS
4. If approved: write `HUMAN_APPROVED` event to Kafka; log to audit; proceed to sourcing
5. If rejected: return to recruiter with feedback; loop until approved
6. On final approval: sync approved requisition to ATS (Greenhouse/Lever/Workday) via connector

Tools:

- `conversational_jd_generator_tool` — GPT-4o call with JD generation prompt
- `skills_taxonomy_lookup_tool` — ESCO/O*NET mapping via Neo4j lookup
- `neo4j_comp_benchmark_tool` — salary percentile query from Talent Intelligence Graph
- `levelsfyi_comp_tool` — Levels.fyi licensed data
- `bls_api_tool` — Bureau of Labor Statistics public API
- `workday_comp_band_tool` — queries company's active comp bands from Workday HCM (if integrated)
- `requisition_health_scorer_tool` — structured analysis returning health score and issues
- `approval_workflow_tool` — creates LangGraph blocking checkpoint; sends notifications
- `ats_sync_tool` — pushes approved requisition to ATS via connector microservice

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2.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
OpenAI API (GPT-4o)	JD generation, health analysis	API Key	Phase 1
Neo4j	Comp benchmarks, skill supply signals	Bolt driver	Phase 1
Levels.fyi	Tech comp benchmarks	Data license	Phase 2
Bureau of Labor Statistics API	Wage statistics	Public API	Phase 1
Workday HCM	Active comp bands	Workday REST API (OAuth 2.0)	Phase 2
Greenhouse REST API	Sync approved requisition to ATS	REST + webhooks	Phase 1
Lever REST API	Sync approved requisition	REST + webhooks	Phase 2
AWS SES	Approval notification emails	AWS SDK	Phase 1

2.4 Technical Components to Build

Component	Type	Tech	Notes
Conversational JD builder UI	Frontend	React 18 + Zustand	Chat-like input + structured JD preview side by side

Component	Type	Tech	Notes
JD editor	Frontend	React 18 + TipTap	Rich text editor; inline comp benchmark widget
Comp benchmark widget	Frontend	React 18 + Recharts	Percentile bar chart; p25/median/p75/p90 display
Requisition health report UI	Frontend	React 18	Score gauge; issue list with fix suggestions
Approval workflow UI	Frontend	React 18	Approval chain status; approve/reject/comment actions
Requisition database schema	Database	PostgreSQL	Requisitions, approval records, skill requirements
ATS connector framework	Backend microservice	Node.js	Generic connector interface; Greenhouse connector first
`integration-hub` service	Backend microservice	Node.js	Token refresh, rate limit compliance, sync scheduling

2.5 Build Complexity

Component	Complexity	Estimated Dev Effort
RequisitionIntelligenceAgent	High	3 weeks (1 AI/ML engineer)
Conversational JD builder UI	Medium	2 weeks (1 FE engineer)
Comp benchmark integration	Medium	1 week (1 backend engineer)
Approval workflow (LangGraph + Kafka)	High	2 weeks (1 backend engineer)
Greenhouse ATS connector	High	2 weeks (1 backend engineer)

3. Module 2 — Candidate Sourcing (RadarAgent)

3.1 Feature Description

RadarAgent discovers candidates across multiple sources — LinkedIn (official API), GitHub, Stack Overflow, internal ATS talent pools, and referral networks. It scores candidates on multiple dimensions (skills match, trajectory alignment, motivation fit, cultural signals) weighted by recruiter priorities. It identifies passive candidates whose public activity signals a potential openness to a move. It delivers a ranked shortlist to the recruiter for approval before any outreach begins.

User-facing outputs:

- Ranked candidate shortlist with multi-dimensional fit score breakdown
- Passive candidate signals surface with evidence (e.g., "Added 3 new skills in last 30 days; following your company page")
- Source attribution for each candidate: how they were found
- Shortlist approval interface: recruiter approves/rejects each candidate before outreach
- Exclusion controls: previously contacted, declined, or blocked candidates filtered automatically

3.2 AI Agents Required

Agent 2.1 — `RadarAgent`

Property	Detail
Role	Multi-source candidate discovery, fit scoring, passive signal detection, shortlist generation
Trigger	Approved requisition enters sourcing phase; also runs on schedule (daily refresh)
LLM Model	Fine-tuned GPT-4o-mini (profile parsing, skill extraction); GPT-4o (motivation and culture signal synthesis)
Orchestration	LangGraph pipeline with parallel multi-source discovery sub-tasks

Tasks:

Multi-Source Discovery:

1. Query LinkedIn Talent Solutions API: boolean skill + location + seniority search
2. Query GitHub API: search users by language expertise, contribution frequency, public activity signals
3. Query Stack Overflow API: users with high reputation scores in required skill tags
4. Query internal ATS talent pool: re-surface previously assessed candidates with suitable profiles
5. Query referral network: flag any connections referred by current employees (via email domain matching)
6. Deduplicate across sources: same person found on LinkedIn + GitHub → merge into single candidate record

Candidate Profiling & Skill Extraction:

1. For each discovered candidate: extract structured profile data (skills, role history, education, location)
2. Map skills to ESCO/O*NET taxonomy
3. Embed candidate profile into Pinecone (user namespace or enterprise namespace)
4. Upsert/update Candidate node in Neo4j Talent Intelligence Graph

Multi-Dimensional Fit Scoring:

1. Compute semantic similarity score: Pinecone cosine similarity between candidate embedding and JD embedding
2. Compute GNN graph match score: GraphSAGE model — queries SageMaker endpoint — incorporates career trajectory context from Neo4j
3. Compute trajectory alignment score: does candidate's career path suggest they're heading toward this role type?
4. Infer motivation fit: use GPT-4o to analyse public writing (GitHub READMEs, Stack Overflow answers, LinkedIn posts) for motivation signals — growth, autonomy, compensation, mission, stability
5. Compute weighted composite score based on recruiter-configured weights per dimension
6. Rank shortlist by composite score

Passive Candidate Signal Detection:

1. Monitor LinkedIn API signals: recent skill additions, company page follow activity
2. Monitor GitHub signals: recent increase in public commit frequency, new repo creation in relevant tech stack
3. Monitor Stack Overflow: recent increase in activity in relevant technology tags
4. Flag candidates with 2+ passive signals as "potentially open to conversations"
5. Surface signal evidence to recruiter: specific, verifiable signals (not vague inferences)

Shortlist Generation & Human Approval Gate:

1. Compile ranked shortlist of top-N candidates (default: 20; configurable)
2. Present to recruiter with full fit score breakdown and evidence per dimension
3. Create `PENDING_HUMAN_APPROVAL` checkpoint (Human Handoff Protocol)
4. Recruiter approves individual candidates for outreach, rejects others, or requests more candidates
5. Approved candidates enter outreach queue; rejected candidates logged (with reason, if given) to audit log

Tools:

- `linkedin_talent_api_tool` — LinkedIn Talent Solutions search API
- `github_user_search_tool` — GitHub REST API: search users by language, topic, activity
- `stackoverflow_user_tool` — Stack Exchange API: users by tag reputation
- `ats_talent_pool_tool` — internal ATS sync: re-surface past candidates
- `referral_match_tool` — email domain matching against company employee list
- `profile_dedup_tool` — cross-source entity resolution (Redis-backed seen-profiles cache)
- `skills_taxonomy_lookup_tool` — ESCO/O*NET mapping
- `pinecone_candidate_upsert_tool` — embed + upsert candidate profile to Pinecone
- `neo4j_candidate_upsert_tool` — upsert Candidate node and skill/career edges to Neo4j
- `pinecone_semantic_match_tool` — cosine similarity match against JD embedding
- `neo4j_gnn_match_tool` — GraphSAGE SageMaker inference endpoint
- `motivation_inference_tool` — GPT-4o call: analyse public writing for motivation signals
- `passive_signal_detector_tool` — rule-based + ML classifier on platform activity signals
- `shortlist_approval_tool` — creates LangGraph blocking checkpoint; sends recruiter notification

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3.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
LinkedIn Talent Solutions API	Candidate search and profile data	OAuth (partner agreement)	Phase 1
GitHub REST API	Developer profile and activity data	OAuth 2.0 / API Key	Phase 1
Stack Exchange API	Technical reputation signals	API Key	Phase 2
Greenhouse REST API	ATS talent pool data	REST + webhooks	Phase 1
Lever REST API	ATS talent pool data	REST + webhooks	Phase 2
Pinecone	Candidate embedding storage and semantic search	API Key	Phase 1
Neo4j	Talent Intelligence Graph — candidate graph, GNN match	Bolt driver	Phase 1
AWS SageMaker	GraphSAGE GNN inference	AWS SDK	Phase 2
Bing Web Search API	Supplementary candidate public content	Azure API Key	Phase 1

3.4 Technical Components to Build

Component	Type	Tech	Notes
Shortlist UI	Frontend	React 18	Ranked candidate cards; fit score breakdown; approve/reject controls
Passive signal indicators	Frontend	React 18	Signal badges on candidate cards with evidence on hover
Source attribution display	Frontend	React 18	"Found via: LinkedIn + GitHub" badges
Candidate deduplication engine	Backend	Python + Redis	Cross-source entity resolution; fuzzy name + email matching
Multi-source crawler scheduler	Backend	Python + AWS EventBridge	Per-requisition scheduled discovery runs
Candidate profile database schema	Database	PostgreSQL	Core candidate records; anonymised ID links to Neo4j

3.5 Build Complexity

Component	Complexity	Estimated Dev Effort
RadarAgent LangGraph pipeline	Very High	4 weeks (1 AI/ML engineer)
LinkedIn Talent Solutions integration	High	2 weeks (1 backend engineer)
GitHub + Stack Overflow integration	Medium	1 week (1 backend engineer)
Multi-dimensional scoring system	High	2 weeks (1 AI/ML engineer)
Passive signal detection	High	2 weeks (1 AI/ML engineer)
Shortlist approval UI	Medium	1 week (1 FE engineer)

4. Module 3 — Candidate Intelligence Brief™ (IntelligenceAgent)

4.1 Feature Description

Automated synthesis of publicly available professional data into a structured narrative brief delivered to the recruiter before every engagement. Every claim in the brief is backed by verifiable evidence and links to its source. Candidates can view what data was used and request corrections (GDPR compliance).

User-facing outputs:

- Career arc narrative: LLM-generated professional trajectory analysis
- Skills evidence map: each claimed skill backed by a specific public evidence item
- Motivation inference: likely priorities based on public signals (not assumptions)
- Conversation recommendations: 3–5 specific, personalised interview angles for this candidate + this role
- Verification flags: gaps, rapid tenure changes, inconsistencies surfaced for recruiter attention
- Data provenance panel: every claim links to its source; transparency for candidate access

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4.2 AI Agents Required

Agent 3.1 — `IntelligenceAgent`

Property	Detail
Role	Master orchestrator for Candidate Intelligence Brief™ generation
Trigger	Candidate approved for outreach by recruiter (shortlist approval gate passed)
LLM Model	GPT-4o (deep multi-source synthesis; long-context narrative generation)
Orchestration	LangGraph pipeline with parallel sub-agent data gathering, followed by synthesis
SLO	Brief must complete in < 45 minutes p95

Tasks (Master Orchestrator Level):

1. Receive trigger: (candidate_id, role_id, recruiter_id)
2. Decompose into 4 parallel data-gathering sub-tasks (see below)
3. Await all sub-agent completions (timeout + partial brief fallback if any source fails)
4. RAG pipeline: chunk, embed, retrieve most relevant context from gathered data
5. Synthesise Candidate Intelligence Brief™ using GPT-4o with full structured prompt
6. Validate output against Pydantic schema (all required sections present and populated)
7. Attach data provenance: link every claim to its source chunk
8. Store brief in PostgreSQL linked to candidate + requisition record
9. Push brief to recruiter dashboard via WebSocket notification
10. Run compliance check: verify no protected characteristic information (age, gender, nationality signals) is explicitly used as a decision signal — `bias-monitor` sidecar validates output before delivery

Sub-tasks:

Agent 3.1a — `LinkedInProfileSubAgent`

Tasks:

1. Fetch full LinkedIn profile via LinkedIn Talent Solutions API
2. Extract: all roles with tenure, company names, job titles, education, skills, endorsements, summary
3. Parse career arc: identify career pivots, progression pattern (linear/multi-functional/specialist), tenure patterns
4. Identify career gaps: periods > 6 months between roles — flag for recruiter awareness
5. Extract recent activity signals: posts, articles, engagement topics

Tools:

- `linkedin_full_profile_fetch_tool` — LinkedIn Talent Solutions API (full profile endpoint)
- `career_arc_parser_tool` — GPT-4o-mini: extracts structured career timeline from LinkedIn data
- `gap_detector_tool` — Python function: identifies tenure gaps in career history
- `linkedin_activity_tool` — LinkedIn API: recent posts and engagement data

Agent 3.1b — `PublicWorkSubAgent`

Tasks:

1. Fetch GitHub: public repos (README, primary language, stars, forks, recent commits), contribution graph, profile bio
2. Fetch GitHub Gists: assess code quality and problem-solving approach

3. Search for publications: academic papers (Google Scholar API), technical blogs (Substack, Medium, Dev.to)
4. Search for talks/presentations: YouTube, Loom, conference archives (SpeakerDeck API)
5. Generate skills evidence map: for each skill claimed on LinkedIn, find a specific public evidence item (GitHub repo, publication, talk) that corroborates it

Tools:

- `github_full_profile_tool` — GitHub REST API: repos, contributions, bio, gists
- `google_scholar_tool` — Google Scholar API (unofficial): publication search by author name
- `medium_substack_search_tool` — Bing search filtered to Medium/Substack/Dev.to
- `youtube_speaker_tool` — YouTube Data API v3: search for talks by name
- `speakerdeck_api_tool` — SpeakerDeck public API: presentation search
- `skills_evidence_mapper_tool` — GPT-4o: maps each skill to its best public evidence item

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Agent 3.1c — `MotivationInferenceSubAgent`

Tasks:

1. Analyse public writing (LinkedIn articles, GitHub READMEs, blog posts, talks) for expressed priorities
2. Analyse job transition patterns: what types of companies/roles has this person moved toward vs. away from?
3. Classify motivation signals across 5 dimensions: growth, autonomy, compensation, mission, stability
4. Assign confidence score to each inferred motivation (high/medium/low based on volume of evidence)
5. Generate motivation narrative: 2–3 sentences explaining likely priorities with evidence citations

Tools:

- `content_sentiment_analyser_tool` — GPT-4o: multi-document sentiment and topic analysis
- `transition_pattern_analyser_tool` — Python function: classifies company types and sizes across career history
- `motivation_classifier_tool` — GPT-4o call with structured motivation classification prompt
- `motivation_confidence_scorer_tool` — scoring function based on evidence volume

Agent 3.1d — `ConversationAngleSubAgent`

Tasks:

1. Cross-reference candidate's profile with the specific role requirements
2. Identify 3–5 high-value interview angles: questions that explore the intersection of candidate's specific experiences and the role's most critical needs
3. Generate at least 1 "verification angle": a question to explore a gap or inconsistency identified in the brief
4. Ensure angles are specific to this candidate — not generic competency questions

Tools:

- `profile_role_crossref_tool` — compares candidate profile against JD requirements
- `interview_angle_generator_tool` — GPT-4o call: generates specific, evidence-grounded interview angles

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4.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
LinkedIn Talent Solutions API	Full candidate profile	OAuth (partner)	Phase 1
GitHub REST API	Public repos, contributions, bio	API Key	Phase 1
Bing Web Search API	Publications, blog posts, articles	Azure API Key	Phase 1
YouTube Data API v3	Talks and presentations	API Key	Phase 2
SpeakerDeck API	Conference presentations	Public API	Phase 2
OpenAI API (GPT-4o, GPT-4o-mini)	Synthesis, classification	API Key	Phase 1
Pinecone	RAG retrieval of candidate data chunks	API Key	Phase 1

Integration	Purpose	Auth Method	Build Phase
Neo4j	Career graph, skill evidence map	Bolt driver	Phase 1

4.4 Technical Components to Build

Component	Type	Tech	Notes
Candidate Intelligence Brief™ UI	Frontend	React 18	Tabbed layout: Career Arc / Skills Evidence / Motivation / Conversation Angles / Verification Flags
Data provenance panel	Frontend	React 18	Click any claim → see source; link to original URL
Brief generation status stream	Frontend	React 18 + WebSocket	Real-time agent progress display
Brief storage schema	Database	PostgreSQL	Brief JSON stored linked to candidate + requisition
Bias monitor sidecar	Backend	Python (containerised sidecar)	Runs alongside agent-orchestrator; validates brief output before delivery
SHAP explainability renderer	Frontend	React 18 + custom visualisation	Match score breakdown with human-readable SHAP values
Candidate data access portal	Frontend	Next.js (candidate-facing)	GDPR: candidates can view their brief data and request corrections

4.5 Build Complexity

Component	Complexity	Estimated Dev Effort
IntelligenceAgent master pipeline	Very High	4 weeks (1 AI/ML engineer)
4 sub-agents	High	3 weeks (1 AI/ML engineer)
Brief UI (tabbed, provenance)	High	2 weeks (1 FE engineer)
Bias monitor sidecar	High	2 weeks (1 backend engineer)
SHAP explainability	High	2 weeks (1 ML engineer)

5. Module 4 — Outreach & Scheduling (OutreachAgent + SchedulerAgent)

5.1 Feature Description

OutreachAgent generates a personalised outreach message for each approved candidate — referencing at least two specific, verifiable signals from their public profile. All messages require recruiter review and approval before sending. OutreachAgent learns from response rates using reinforcement learning on response signals, continuously improving message framing and timing. SchedulerAgent handles the full interview scheduling lifecycle: calendar negotiation, confirmation with pre-meeting briefs, and autonomous rescheduling.

User-facing outputs:

- Personalised message drafts for each approved candidate (editable before send)
- Send queue with configurable timing (e.g., send Tuesday 9–11am for highest response rates)
- Response rate analytics per message variant
- Calendar availability negotiation status per candidate
- Interview confirmation with context-rich pre-meeting brief attached
- Rescheduling handled automatically with recruiter notification

5.2 AI Agents Required

Agent 4.1 — `OutreachAgent`

Property	Detail

Property	Detail
Role	Generates personalised candidate outreach messages; manages send queue; learns from response rates
Trigger	Recruiter approves shortlisted candidate for outreach
LLM Model	GPT-4o-mini (cost-efficient; high volume; quality monitored via A/B response rate data)
Orchestration	LangGraph task triggered by approval event; A/B testing loop managed by orchestrator

Tasks:

Personalised Message Generation:

1. Retrieve Candidate Intelligence Brief™ for this candidate (pre-generated in Module 3)
2. Extract 2–3 strongest specific signals from brief: concrete, verifiable items (a specific GitHub project, a blog post, a career pivot)
3. Retrieve role context: company, role title, key selling points (growth stage, team, impact)
4. Generate personalised message: opens with specific candidate signal, frames role relevance, clear call to action
5. Style the message per recruiter's configured tone (formal/conversational/technical)
6. Return message draft to recruiter for review — message cannot send without recruiter approval

Approval Gate:

1. Create `PENDING_HUMAN_APPROVAL` checkpoint for each message
2. Recruiter reviews draft, edits if needed, approves
3. Approved message enters send queue

Send Queue Management:

1. Schedule send at optimal time: research-backed default (Tuesday–Thursday, 9–11am local time of candidate) or recruiter-configured
2. Deduplication check: Redis cache — has this candidate been contacted for this role within the past 90 days?
3. Execute send via AWS SES (email) or LinkedIn InMail API (if recruiter prefers InMail)
4. Log send event to audit log

Response Rate Learning:

1. Track response/no-response per message variant (different signal types, different opening styles, different CTAs)
2. Feed response signals back to OutreachAgent via Kafka event stream
3. Use Thompson Sampling (simple Bayesian bandit) to adjust message variant weights over time
4. Surface response rate analytics to recruiter dashboard

Tools:

- `brief_retriever_tool` — fetches Candidate Intelligence Brief™ from PostgreSQL
- `signal_extractor_tool` — GPT-4o-mini: identifies 2–3 strongest specific signals from brief
- `message_generator_tool` — GPT-4o-mini: generates personalised message with specific signal references
- `message_approval_tool` — LangGraph blocking checkpoint; recruiter notification
- `outreach_dedup_tool` — Redis: checks if candidate has been contacted recently
- `ses_send_tool` — AWS SES email delivery
- `linkedin_inmail_tool` — LinkedIn Recruiter InMail API (if enabled)
- `response_tracker_tool` — tracks reply/no-reply per message; feeds into bandit model
- `thompson_sampling_tool` — Bayesian bandit for message variant optimisation

Agent 4.2 — `SchedulerAgent`

Property	Detail
Role	Negotiates interview availability; books interviews; manages rescheduling
Trigger	Candidate responds positively to outreach and indicates availability
LLM Model	GPT-4o-mini (availability parsing from candidate email/message)
Orchestration	LangGraph event-driven state machine

Tasks:

1. Parse candidate's availability response from email or in-app message
2. Query recruiter's calendar (Google Calendar or Outlook) for available slots via free/busy API
3. Propose 3 interview time options within candidate's stated availability window
4. Candidate selects a slot (via embedded scheduling link or CareerAgent™ calendar)
5. Create calendar event for both recruiter and candidate
6. Send confirmation to both parties: includes context-rich pre-meeting brief (link to sanitised Candidate Intelligence Brief™ for recruiter; Opportunity Intelligence Brief™ for candidate via CareerAgent™)
7. Handle rescheduling: if candidate or recruiter cancels, propose new slots automatically — notify recruiter of change
8. Handle no-show: flag to recruiter 5 minutes after scheduled start if interview hasn't been joined

Tools:

- `email_availability_parser_tool` — GPT-4o-mini: extracts availability windows from free text
- `google_calendar_tool` — Google Calendar API: free/busy query + event creation
- `outlook_calendar_tool` — Microsoft Graph API: free/busy query + event creation
- `calendly_fallback_tool` — Calendly API: alternative scheduling flow if recruiter prefers
- `interview_confirmation_mailer_tool` — AWS SES: sends confirmation with brief links
- `reschedule_manager_tool` — handles cancellation + reproposal flow
- `no_show_alerter_tool` — time-based event trigger (AWS EventBridge) if interview not joined

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5.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
OpenAI API (GPT-4o-mini)	Message generation, availability parsing	API Key	Phase 1
AWS SES	Email outreach delivery + confirmations	AWS SDK	Phase 1
LinkedIn Recruiter InMail API	InMail outreach (optional recruiter preference)	LinkedIn partner OAuth	Phase 2
Google Calendar API	Recruiter + candidate calendar scheduling	OAuth 2.0	Phase 1
Microsoft Graph API (Outlook)	Same for Microsoft 365 customers	OAuth 2.0 (Azure AD app)	Phase 1
Calendly API	Alternative scheduling flow	OAuth 2.0	Phase 2
Redis	Outreach deduplication	AWS ElastiCache	Phase 1
AWS EventBridge	Scheduled actions (optimal send time, no-show alert)	AWS SDK	Phase 1

5.4 Technical Components to Build

Component	Type	Tech	Notes
Outreach message editor UI	Frontend	React 18 + TipTap	Pre-populated with agent draft; recruiter edits inline
Send queue dashboard	Frontend	React 18	Queue items with status; scheduled send time; approve/reject
Response rate analytics panel	Frontend	React 18 + Recharts	Per message variant; send/response rates; trend chart
Scheduling status tracker	Frontend	React 18	Per candidate: scheduling state; calendar event link
Calendar OAuth flows	Backend	Node.js + Cognito	Google + Microsoft OAuth tokens stored in Secrets Manager
Scheduling confirmation emails	Backend	Node.js + SES templates	HTML email templates with brief links
No-show alerter	Backend	Python + EventBridge	Time-triggered lambda: check if interview started

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5.5 Build Complexity

Component	Complexity	Estimated Dev Effort

Component	Complexity	Estimated Dev Effort
OutreachAgent (message gen + bandit learning)	High	3 weeks (1 AI/ML engineer)
SchedulerAgent (multi-calendar)	High	3 weeks (1 backend engineer)
Google Calendar + Outlook OAuth	Medium	1 week (1 backend engineer)
Outreach editor UI	Medium	1 week (1 FE engineer)
Response rate analytics	Medium	1 week (1 FE + 1 backend engineer)

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6. Module 5 — Screen & Assessment (ScreenAgent)

6.1 Feature Description

Async video screening with AI-generated role-specific questions. Responses analysed for content quality, communication clarity, and structured competency signals. Portfolio and work-sample analysis for technical roles. Structured capability profile output fed into the Candidate Intelligence Brief™ as a verified competency layer.

User-facing outputs (recruiter):

- Async video screening platform embedded in HOGH dashboard
- AI-generated screening questions specific to this role and candidate
- Transcript + annotated analysis per response: content score, communication clarity, competency signals
- Structured capability profile added to Candidate Intelligence Brief™
- Comparative scoring across all screened candidates for this requisition

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6.2 AI Agents Required

Agent 5.1 — `ScreenAgent`

Property	Detail
Role	Generates screening questions; analyses video responses; assesses portfolios and work samples
Trigger	Recruiter advances candidate to screening stage in pipeline
LLM Model	GPT-4o (question generation, response analysis, capability profile synthesis)
Orchestration	LangGraph pipeline: question generation → candidate submission → analysis

Tasks:

Question Generation:

1. Retrieve role JD and Candidate Intelligence Brief™
2. Generate 4–6 role-specific screening questions:
    - 2 competency questions targeting the role's must-have skills (from JD)
    - 1–2 behavioural questions targeting motivations identified in brief
    - 1 "gap exploration" question targeting a verification flag in the brief
3. Return questions to recruiter for review/edit before sending to candidate

Video Response Analysis:

1. Receive video response file (uploaded by candidate via screening platform)
2. Transcribe video: AWS Transcribe (batch mode; async)
3. Analyse transcription per question:
    - Content quality: did the answer address the question? Is the claim specific and credible?
    - STAR structure: situation, task, action, result — all present?
    - Communication clarity: filler word rate, pacing, vocabulary complexity
    - Competency signals: extract specific claims → map to ESCO skill taxonomy
4. Generate competency signal per answer: "Candidate provided specific evidence of Python expertise at scale (claim: 'led migration of 50M row ETL pipeline from Pandas to PySpark') — rated: Strong"

5. Generate capability profile: structured JSON summarising all competency signals with confidence ratings

Portfolio Analysis (Technical Roles):

1. Receive GitHub repo URL or portfolio link from candidate profile
2. Analyse primary repo(s): code quality signals (test coverage, documentation, architecture patterns), technology stack, commit history and consistency
3. Generate portfolio assessment: 3–5 specific findings with evidence

Structured Capability Profile:

1. Synthesise video analysis + portfolio assessment into unified Structured Capability Profile
2. Add Structured Capability Profile as a new section to the existing Candidate Intelligence Brief™
3. Update candidate's Neo4j profile: add evidence-backed `HAS_SKILL` edges with `evidence_count` from screen

Tools:

- `brief_retriever_tool` — fetches brief from PostgreSQL
- `screen_question_generator_tool` — GPT-4o: generates role-specific screening questions
- `video_transcription_tool` — AWS Transcribe batch: transcribes video response
- `response_analyser_tool` — GPT-4o: structured analysis of transcribed response
- `star_structure_scorer_tool` — GPT-4o-mini: STAR adherence scoring
- `competency_signal_extractor_tool` — fine-tuned GPT-4o-mini: maps answer content to ESCO skills
- `github_portfolio_analyser_tool` — GitHub REST API + GPT-4o: code quality and pattern analysis
- `capability_profile_generator_tool` — GPT-4o: synthesises all signals into structured profile
- `neo4j_skill_evidence_updater_tool` — upserts `HAS_SKILL` edges with screen evidence

6.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
AWS Transcribe (batch)	Video response transcription	AWS SDK	Phase 1
AWS S3	Video response file storage	AWS SDK	Phase 1
OpenAI API (GPT-4o, GPT-4o-mini)	Question gen, response analysis, portfolio analysis	API Key	Phase 1
GitHub REST API	Portfolio and code sample analysis	API Key	Phase 1
Neo4j	Update candidate skill evidence	Bolt driver	Phase 1
Zoom API or **daily.co API**	Async video recording infrastructure (if not using self-hosted)	API Key	Phase 1

6.4 Technical Components to Build

Component	Type	Tech	Notes
Async video recording UI	Frontend	React 18 + WebRTC	Embedded in candidate-facing screen invite flow
Video upload + S3 storage pipeline	Backend	Node.js + AWS S3 + SQS	Trigger ScreenAgent on upload completion
Screen analysis results UI (recruiter)	Frontend	React 18	Per-question scores, transcript, capability profile panel
Comparative scoring UI	Frontend	React 18 + Recharts	Cross-candidate capability comparison chart
Candidate screening invite flow	Frontend	React 18 + email	Candidate receives invite, records async, submits

---

6.5 Build Complexity

Component	Complexity	Estimated Dev Effort
ScreenAgent (full pipeline)	Very High	4 weeks (1 AI/ML engineer)

Component	Complexity	Estimated Dev Effort
Video recording infrastructure	High	2 weeks (1 backend + 1 FE engineer)
Transcription + analysis pipeline	High	2 weeks (1 backend engineer)
Portfolio analysis agent	High	2 weeks (1 AI/ML engineer)
Screen results UI	Medium	2 weeks (1 FE engineer)

---

7. Module 6 — Interview Orchestration

7.1 Feature Description

Pre-interview pack delivered automatically to the recruiter: final Candidate Intelligence Brief™ with updated screen data, recommended questions, compensation context. Video interview hosted in-platform or via Zoom/Teams. Post-interview debrief prompt sent immediately after the call. Feedback used to update candidate ranking. CareerAgent notifies the job seeker of outcome and provides anonymised feedback where recruiter has enabled this.

7.2 AI Agents Required

Agent 6.1 — `InterviewOrchestrationAgent`

Property	Detail
Role	Manages pre-interview pack delivery, video hosting, post-interview debrief collection
Trigger	Interview confirmed and scheduled (SchedulerAgent event)
LLM Model	GPT-4o (debrief synthesis, feedback summary)
Orchestration	LangGraph event-driven state machine keyed to interview event

Tasks:

Pre-Interview Pack:

1. 24 hours before interview: compile final Candidate Intelligence Brief™ (with screen data added)
2. Add recommended interview questions: from IntelligenceAgent conversation angles + ScreenAgent gap exploration
3. Add compensation context: current benchmark vs. candidate's inferred comp expectations (from CareerAgent data if candidate has shared)
4. Push to recruiter dashboard: WebSocket notification + email summary
5. Push Opportunity Intelligence Brief™ to candidate via CareerAgent™ (if CareerAgent user)

Video Interview Hosting:

1. If in-platform hosting: generate HOGH video room (WebRTC-based) with in-interview tools (note-taking, competency scoring checklist, question list)
2. If Zoom: create Zoom meeting via Zoom API; send join links to both parties
3. If Teams: create Teams meeting via Microsoft Graph API; send join links
4. Monitor meeting join events: detect when both parties have joined

Post-Interview Debrief:

1. 10 minutes after scheduled interview end time: push debrief prompt to recruiter
2. Debrief template: structured feedback form — competency ratings (per required skill), culture fit signal, communication effectiveness, hire/no-hire recommendation with reasoning
3. Recruiter completes debrief in HOGH or integrated ATS
4. Store debrief linked to candidate + requisition record
5. Update candidate composite score: weight debrief ratings into overall candidate ranking
6. Generate feedback summary for candidate (anonymised per recruiter preference):
    - If recruiter enables feedback: CareerAgent™ receives structured feedback for candidate
    - If recruiter disables: candidate notified of outcome only

Outcome Communication:

1. Recruiter marks decision: Advance / Reject / Hold
2. If Reject: CareerAgent™ notified; anonymous feedback delivered (if enabled) within 24 hours
3. If Advance: candidate moves to next stage; SchedulerAgent triggered for next interview booking
4. All decisions and timestamps written to audit log

Tools:

- `pre_interview_pack_generator_tool` — assembles updated brief + questions + comp context
- `zoom_meeting_creator_tool` — Zoom API: create meeting + send links
- `teams_meeting_creator_tool` — Microsoft Graph API: create meeting
- `webrtc_room_tool` — internal video room creation (daily.co API or self-hosted Mediasoup)
- `debrief_prompt_sender_tool` — EventBridge-triggered push at T+10min post-interview
- `debrief_collector_tool` — structured form submission → PostgreSQL
- `candidate_score_updater_tool` — updates candidate composite score with debrief ratings
- `outcome_communicator_tool` — triggers CareerAgent™ notification via Kafka event
- `audit_log_writer_tool` — immutable write to audit log

7.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
Zoom API	Video interview creation	OAuth 2.0	Phase 1
Microsoft Graph API (Teams)	Teams meeting creation	OAuth 2.0 (Azure AD)	Phase 1
daily.co API or **Mediasoup**	In-platform video rooms	API Key / self-hosted	Phase 2
AWS EventBridge	Debrief prompt timing	AWS SDK	Phase 1
OpenAI API (GPT-4o)	Debrief synthesis, feedback summary	API Key	Phase 1

7.4 Build Complexity

Component	Complexity	Estimated Dev Effort
Pre-interview pack compilation	Medium	1 week (1 AI/ML engineer)
Zoom + Teams integration	Medium	2 weeks (1 backend engineer)
In-platform video rooms	Very High	4 weeks (1 backend + 1 FE engineer) — Phase 2
Post-interview debrief flow	Medium	1 week (1 FE + 1 backend engineer)
Outcome → CareerAgent communication	Medium	1 week (1 backend engineer)

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8. Module 7 — Offer Management (OfferAgent)

8.1 Feature Description

OfferAgent generates a recommended offer package benchmarked against the Talent Intelligence Graph and the company's own historical offer distribution. Models compensation scenarios ("if equity cliff extended, cash equivalent?"). Every offer requires human recruiter approval before generation. Offer letter generated only after verbal agreement reached. NegotiationCoach™ activates for the live offer conversation.

Critical compliance constraint: The `offer-service` database has a hard constraint — an offer letter cannot be generated unless the associated `offer_approval_id` references a confirmed human approval record in the audit log. This is enforced at the database level, not application code.

8.2 AI Agents Required

Agent 7.1 — `OfferAgent`

Property	Detail
Role	Generates recommended offer package; models comp scenarios; manages offer approval workflow
Trigger	Recruiter initiates offer process for a candidate
LLM Model	GPT-4o (scenario reasoning, offer letter generation)
Orchestration	LangGraph state machine with multiple human approval gates

Tasks:

Offer Package Generation:

1. Retrieve role's approved comp band from Workday HCM / HR system
2. Query Neo4j: market benchmark for this role + location + seniority (p25/median/p75/p90)
3. Query platform aggregate data: company's own historical offer distribution (anonymised) — what has this company typically offered for this role type?
4. Retrieve candidate's inferred compensation expectations (from CareerAgent data if candidate has shared publicly; surfaced as a range, not a precise figure)
5. Generate recommended offer package:
    - Base salary recommendation: specific figure within approved comp band, anchored to market data
    - Bonus recommendation: percentage and structure
    - Equity recommendation: grant size, cliff, vesting schedule
    - Benefits package: standard + any flex benefits available
    - Signing bonus (if applicable): recommendation with rationale

Scenario Modelling:

1. Model "comp equivalent" scenarios: "if we extend the equity cliff by 6 months, what's the cash equivalent for the candidate?"
2. Model "trade-off" scenarios: "if candidate requests +$15K base, what's the total comp impact over 3 years vs. offering more equity?"
3. Generate scenario modelling table for recruiter review
4. Identify guardrails: flag if any scenario pushes outside the approved comp band

Approval Workflow:

1. Present recommended offer to recruiter for review
2. Create `PENDING_HUMAN_APPROVAL` checkpoint (Human Handoff Protocol)
3. Recruiter approves/adjusts and approves — approval record written to audit log
4. On recruiter approval: offer details locked; `offer_approval_id` generated and stored
5. Offer is in "verbal discussion" state — no letter generated yet

Offer Letter Generation:

1. Triggered only after recruiter confirms verbal agreement with candidate
2. Check database constraint: `offer_approval_id` must exist and be confirmed
3. Generate offer letter from approved offer details (using legal-compliant template)
4. Route offer letter to HR Ops for counter-signature (approval workflow)
5. On final signature: deliver offer letter to candidate via CareerAgent™ + email

Tools:

- `workday_comp_band_tool` — active comp bands from Workday HCM
- `neo4j_market_benchmark_tool` — salary benchmarks from Talent Intelligence Graph
- `platform_offer_history_tool` — anonymised company historical offer distribution
- `career_agent_inferred_comp_tool` — candidate's shared comp expectations (anonymised range)
- `offer_package_generator_tool` — GPT-4o: generates recommended offer with rationale
- `scenario_modeller_tool` — Python function + GPT-4o: comp scenario calculations and narratives
- `offer_approval_tool` — LangGraph blocking checkpoint; creates offer_approval_id on confirmation
- `offer_letter_generator_tool` — GPT-4o + legal template: generates offer letter document
- `offer_letter_delivery_tool` — SES + CareerAgent™ Kafka event: delivers offer to candidate
- `audit_log_writer_tool` — immutable write for every offer action

8.3 Integrations Required

Integration	Purpose	Auth Method	Build Phase
Workday HCM	Active comp bands, employee records	Workday REST API (OAuth 2.0)	Phase 2
Neo4j	Market benchmarks	Bolt driver	Phase 1
AWS SES	Offer letter delivery	AWS SDK	Phase 1
OpenAI API (GPT-4o)	Offer package generation, scenario reasoning, offer letter	API Key	Phase 1
DocuSign API (optional)	Digital offer letter signature	OAuth 2.0	Phase 2

8.4 Technical Components to Build

Component	Type	Tech	Notes
Offer builder UI	Frontend	React 18	Recommended offer display; scenario modelling table; edit controls
Scenario modelling UI	Frontend	React 18 + Recharts	Interactive scenario explorer: adjust comp components, see total comp impact
Offer approval flow UI	Frontend	React 18	Approval chain tracker; approve/reject buttons
Offer letter template	Backend	Node.js + PDF generation	Legal-compliant template with variable fields
`offer-service` microservice	Backend	Node.js + PostgreSQL	Core offer CRUD; database-level constraint on offer letter generation
Offer approval ID constraint	Database	PostgreSQL	NOT NULL FK constraint on `offer_approval_id` — enforced at DB level

8.5 Build Complexity

Component	Complexity	Estimated Dev Effort
OfferAgent LangGraph pipeline	High	3 weeks (1 AI/ML engineer)
Scenario modelling UI	High	2 weeks (1 FE engineer)
Database-level offer constraint	Medium	0.5 weeks (1 backend engineer)
Offer letter generation + delivery	Medium	1 week (1 backend engineer)
Workday comp band integration	High	2 weeks (1 backend engineer)

9. Module 8 — Analytics & Reporting

9.1 Feature Description

Real-time pipeline velocity dashboard. Diversity analytics with bias audit results. Recruiter performance metrics benchmarked against platform averages. CHRO-level auto-generated reports exportable as PDF or pushed to HRIS.

User-facing outputs:

- Pipeline velocity dashboard: all requisitions by stage, time-in-stage, predicted time-to-fill
- Diversity funnel: demographic composition at each pipeline stage; bias audit results
- Recruiter performance scorecard: conversion rates, response rates, offer acceptance rates
- CHRO board report: auto-generated PDF; pushed to Workday/SuccessFactors on schedule

9.2 AI Agents Required

Agent 8.1 — `AnalyticsAgent`

Property	Detail
Role	Aggregates pipeline data; generates CHRO reports; surfaces recruiter performance insights
Trigger	Scheduled (daily/weekly CHRO report); real-time for pipeline dashboard
LLM Model	GPT-4o-mini (narrative generation for CHRO report text)
Orchestration	Scheduled EventBridge job for reports; real-time WebSocket for dashboard

Tasks:

1. Aggregate pipeline metrics from `recruiter-service` PostgreSQL: stage counts, time-in-stage, conversion rates per stage
2. Compute predicted time-to-fill: regression model based on current stage counts, historical time-to-fill, and pipeline velocity
3. Aggregate diversity data: stage-by-stage demographic composition (using proxy signals where direct data unavailable)
4. Run disparate impact tests per pipeline stage (80% rule across gender, age group, ethnicity)
5. Compute recruiter performance metrics: outreach response rate, scheduling conversion, offer acceptance rate
6. Benchmark against platform anonymised averages: "your offer acceptance rate is 72% vs. platform average 68%"
7. Generate CHRO report narrative: GPT-4o-mini call; produces executive-level summary with data
8. Render CHRO report to PDF (WeasyPrint)
9. Push to Workday or SuccessFactors via HRIS integration

Tools:

- `pipeline_metrics_aggregator_tool` — SQL queries against `recruiter-service` PostgreSQL
- `time_to_fill_predictor_tool` — statistical regression model (Python scikit-learn, hosted on Lambda)
- `diversity_funnel_calculator_tool` — demographic aggregation + disparate impact test
- `platform_benchmark_tool` — queries anonymised platform aggregate data (PostgreSQL + Athena)
- `chro_report_generator_tool` — GPT-4o-mini: generates report narrative from structured data
- `pdf_renderer_tool` — WeasyPrint: renders CHRO report to PDF
- `hris_push_tool` — pushes report to Workday / SuccessFactors via HRIS integration

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9.3 Technical Components to Build

Component	Type	Tech	Notes
Pipeline velocity dashboard	Frontend	React 18 + Recharts	Real-time Kanban + funnel chart; stage velocity metrics
Diversity analytics dashboard	Frontend	React 18 + Recharts	Stage-by-stage demographic funnel; bias flag alerts
Recruiter performance scorecard	Frontend	React 18 + Recharts	Per-metric gauge + trend line; platform benchmark overlay
CHRO report builder	Frontend	React 18	Preview, date range selector, export to PDF
Analytics data warehouse	Backend	AWS Athena + S3 (Parquet)	Kafka → Kinesis Firehose → S3; queried via Athena
`analytics-service` microservice	Backend	Python + FastAPI	Data aggregation, metric computation, Athena query layer

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10. Module 9 — NegotiationCoach™ Live Session (Recruiter Side)

10.1 Feature Description

During the live offer conversation, NegotiationCoach™ delivers real-time private advisory to the recruiter through the TalentAgent™ dashboard. The recruiter sees: live offer comparison vs. market benchmarks and their own comp percentiles, candidate's inferred compensation floor (as a range, never a precise figure), candidate's shared priority stack (only if candidate has enabled this in their CareerAgent™ privacy settings), competing offer signals, and if-then guidance updated as the conversation progresses.

Activation: Both recruiter and candidate must explicitly acknowledge that NegotiationCoach™ is active before the session begins. This is a mutual acknowledgment — recorded and logged.

10.2 AI Agents Required

Agent 9.1 — `NegotiationCoachRecruiterAgent`

Property	Detail
Role	Delivers real-time private negotiation advisory to recruiter during live offer conversation
Trigger	Both parties acknowledge NegotiationCoach™ activation
LLM Model	GPT-4o (real-time game-theoretic scenario reasoning; < 800ms p95 advisory latency)
Orchestration	Persistent WebSocket session; event-driven update loop

Tasks:

1. Pre-session: load offer data, market benchmarks, candidate inferred comp floor (range), candidate's priority stack (if shared)
2. Load company's approved comp band guardrails — NegotiationCoach™ will never suggest offers outside these bands
3. Establish private WebSocket channel for recruiter (separate from candidate channel — zero data sharing)
4. Display: offer vs. market p25/50/75; offer vs. company's p25/50/75 historical distribution; candidate's inferred floor range
5. Accept real-time input: recruiter types "candidate said X" or "candidate is pushing back on equity"
6. Generate if-then guidance in real-time:
   - Candidate pushes back on base → "Consider extending equity cliff by 6 months. Cash equivalent: +$8K over 2 years."
   - Candidate mentions competing offer → "Candidate has leverage. Recommend going to 65th percentile on total comp."
   - Candidate asks about remote flexibility → "Remote is configurable for this band per HR policy."
7. Policy guardrails: flag if recruiter is about to deviate from approved comp band
8. Session end: generate negotiation debrief — agreed terms, ZOPA analysis, recommendations for future offers at this company level

Tools:

- `websocket_recruiter_stream_tool` — private WebSocket channel (recruiter side only)
- `offer_benchmark_display_tool` — real-time benchmark vs. market and company distribution
- `candidate_floor_display_tool` — candidate's inferred comp floor as anonymised range (from CareerAgent™ shared signals only if candidate opted in)
- `comp_band_guardrail_tool` — checks scenario against approved comp band; flags violations
- `scenario_advisor_recruiter_tool` — GPT-4o: if-then recommendation given current conversation state
- `negotiation_debrief_tool` — GPT-4o: post-session summary and ZOPA analysis
- `audit_log_writer_tool` — immutable log of every session event

10.3 Technical Components to Build

Component	Type	Notes
NegotiationCoach™ session UI (recruiter)	React 18	Private panel: benchmark display, candidate signals, coaching prompts, input field
Private WebSocket channel management	Python + FastAPI + AWS API Gateway	Strict isolation — recruiter and candidate channels never share data
Mutual acknowledgment gate	React 18 + PostgreSQL	Both-party acknowledgment logged before session activates
Comp band guardrail enforcement	Python	Hard block if suggestion would violate approved band
Session audit log	PostgreSQL (append-only)	Every session event timestamped and immutable

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11. Module 10 — Compliance & Bias Architecture

11.1 Overview

Compliance and bias monitoring are cross-cutting concerns — they intercept agent outputs at multiple pipeline stages. This module is not a feature presented to recruiters directly; it is the trust infrastructure that makes the platform legally and ethically defensible.

11.2 AI Agents Required

Agent 10.1 — `BiasMonitorAgent`

Property	Detail
Role	Runs disparate impact tests on shortlists and rankings; flags bias; generates corrective re-rankings
Trigger	Every candidate shortlist or ranking update (runs as sidecar alongside `agent-orchestrator`)
LLM Model	Claude 3.5 Sonnet (Constitutional AI training for fairness-sensitive reasoning)
Orchestration	Sidecar container alongside `agent-orchestrator`; intercepts pipeline at defined checkpoints

Tasks:

1. Receive current candidate shortlist or ranking from `agent-orchestrator`
2. Run **disparate impact test** (80% rule / four-fifths rule) across gender, age group, and ethnicity using proxy signals (names, educational institution names, graduation years as age proxy)
3. If shortlist passes: approve and forward to recruiter
4. If shortlist fails:
   - Pause pipeline; write `BIAS_FLAG` event to Kafka and immutable audit log
   - Run SHAP explainability: which features are driving the demographic imbalance?
   - Generate corrective re-ranked shortlist: re-weight scores with bias-corrective factor
   - Generate explainability report for recruiter: human-readable SHAP breakdown
   - Present to recruiter: original shortlist, bias flag, corrective shortlist, and explanation
5. Enterprise HR Ops can configure disparate impact threshold (within EEOC-compliant bounds) — cannot disable monitor entirely

Tools:

- `disparate_impact_tester_tool` — Python statistical function: 80% rule test on shortlist
- `proxy_demographic_classifier_tool` — ML classifier: infers demographic proxy signals from name + education institution data (US Census name-to-ethnicity models)
- `shap_explainer_tool` — SHAP value computation on candidate match score features
- `corrective_ranker_tool` — re-weighting algorithm that reduces demographic skew while preserving skill-based ordering
- `bias_report_generator_tool` — Claude 3.5 Sonnet: generates human-readable bias report
- `audit_log_writer_tool` — immutable write of `BIAS_FLAG` event

Agent 10.2 — `ComplianceAgent`

Property	Detail
Role	Processes DSAR requests; generates compliance reports; monitors EU AI Act obligations
Trigger	DSAR request submitted by candidate; scheduled compliance report (monthly); EU AI Act audit event
LLM Model	Claude 3.5 Sonnet (compliance-sensitive reasoning)
Orchestration	Event-driven Lambda functions for DSAR; scheduled EventBridge for reports

Tasks:

1. **DSAR Processing:**
   - Receive DSAR request from candidate (via HOGH self-service portal or email)
   - Export all data held about candidate: PostgreSQL records, Pinecone vectors (fetched by metadata filter), Neo4j anonymised node data
   - Generate structured data export package (JSON + PDF readable format)
   - Process deletion request: cascade deletion across PostgreSQL, Pinecone (vector deletion by metadata filter), Neo4j (PII field clearing while preserving relationship structure)
   - Log all DSAR actions to immutable audit log with timestamp and processing agent

2. **Compliance Reporting:**
   - Monthly: generate audit report for HR Ops — all AI-assisted decisions, human approval events, bias flags, SHAP explanations generated
   - Export to PDF; available in HOGH Admin Console
3. **EU AI Act Monitoring:**
   - Track completeness of required technical documentation
   - Generate transparency disclosures for candidates (what AI was used, what data was processed)
   - Monitor model drift alerts (flag if model outputs deviate significantly from baseline)

Tools:

- `dsar_data_exporter_tool` — queries all data sources for a given candidate_id
- `pinecone_vector_deletion_tool` — deletes vectors by metadata filter (candidate_id)
- `neo4j_pii_clearer_tool` — clears PII fields on Candidate node while preserving relationship edges
- `postgresql_soft_delete_tool` — sets is_deleted flag; hard delete only via DSAR flow
- `compliance_report_generator_tool` — Claude 3.5 Sonnet: generates audit narrative from structured log data
- `eu_ai_act_disclosure_tool` — generates per-candidate AI usage disclosure for transparency portal

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11.3 Technical Components to Build

Component	Type	Tech	Notes
`bias-monitor` sidecar container	Backend	Python (Docker sidecar)	Runs alongside agent-orchestrator pod in Kubernetes
SHAP explainability computation	Backend	Python (SHAP library)	scikit-learn SHAP on match score features
Bias flag + corrective shortlist UI	Frontend	React 18	Alert banner; original vs. corrective shortlist toggle
DSAR self-service portal	Frontend	Next.js (candidate-facing)	Request export, view data, request deletion
`compliance-service` microservice	Backend	Python + FastAPI + Lambda	DSAR processing, compliance reports, EU AI Act monitoring
Immutable audit log	Database	PostgreSQL (append-only; RLS; row-level delete disabled)	All compliance-critical events
Audit log archive	Storage	AWS S3 (MFA-delete protected; 7-year retention)	Cold archive of audit log exports

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12. Module 11 — HOGH Admin Console & Multi-Tenant Architecture

12.1 Feature Description

The HOGH Admin Console is used by enterprise HR Ops and compliance officers. It provides: user management (add/remove recruiters, configure roles), ATS integration configuration, per-customer NegotiationCoach™ enablement (feature flag), bias monitor threshold configuration, compliance report access, and audit log access.

12.2 Technical Components to Build

Component	Type	Tech	Notes
Admin Console UI	Frontend	React 18	Enterprise-grade admin panel; accessible via SSO
Enterprise SSO configuration	Backend	AWS Cognito + SAML 2.0 / OIDC	Per-customer IdP configuration (Okta, Azure AD, Google Workspace)
RBAC configuration UI	Frontend	React 18	Role assignment: recruiter, hiring_manager, hr_admin, compliance_officer

Component	Type	Tech	Notes
ATS integration config UI	Frontend	React 18	Connect/disconnect ATS; configure sync settings; test connection
Feature flag management	Backend	LaunchDarkly	NegotiationCoach™ gated per enterprise customer; recruiter-level flag management
Bias threshold configuration	Backend	PostgreSQL + UI	Configurable disparate impact threshold per customer (within EEOC bounds)
Prompt customisation (enterprise)	Backend	PostgreSQL `prompt-registry`	Enterprise customers can configure IntelligenceAgent emphasis
Multi-tenant data isolation	Database	PostgreSQL RLS + per-tenant Pinecone namespaces	Per-customer row-level security

13. Cross-Cutting Infrastructure

13.1 `recruiter-service` — Core Backend Service

The central service managing all recruiter-side data.

Responsibilities:

- Recruiter user management, RBAC, permissions
- Requisition CRUD, pipeline stage management
- Candidate records (HOGH-enriched; ATS is source of record for HR compliance data)
- Human Handoff Protocol checkpoint management
- Interview scheduling records
- Row-level security: per-tenant data isolation

Tech: Node.js + Express + PostgreSQL (Aurora RDS, Multi-AZ)

13.2 `agent-orchestrator` — Master Orchestrator Service

The cognitive backbone of TalentAgent™.

Orchestrator loop for a single requisition:

```
1. RECEIVE goal: "Fill ML Engineer role, London, £120K, 45 days"
2. DECOMPOSE into sub-tasks
3. ASSIGN sub-tasks to specialist sub-agents via Kafka task queue
4. MONITOR sub-agent progress via Kafka event stream
5. EVALUATE outputs — re-queue if quality threshold not met (max 3 retries)
6. CHECK Human Handoff Protocol at each defined gate
   → PAUSE if human approval required → NOTIFY → AWAIT
7. SYNTHESISE outputs into deliverables
8. REPORT progress via WebSocket to recruiter dashboard
9. ADAPT plan dynamically (low response rates → trigger OutreachAgent strategy revision)
```

State management: LangGraph persistent state graph per requisition. Checkpointed to PostgreSQL every 60 seconds and on every human approval event. Full replay capability for audit.

Tech: Python + FastAPI + LangGraph + Kafka (AWS MSK)

13.3 Human Handoff Protocol™ — Implementation

Every human-required gate in TalentAgent™:

Gate	Trigger	Action Without Human Approval

Gate	Trigger	Action Without Human Approval
Requisition approval	New requisition created	Sourcing cannot begin
Shortlist approval	RadarAgent generates shortlist	No outreach can be initiated
Outreach approval	OutreachAgent drafts message	Message cannot send
Screen question approval	ScreenAgent generates questions	Screening cannot launch
Offer approval	OfferAgent generates recommendation	Offer letter cannot be generated (DB constraint)
Offer letter approval	Verbal agreement reached	Letter not delivered to candidate

All gates are implemented as **blocking async checkpoints** in LangGraph. The `offer-service` has an additional **database-level constraint** (FK to audit-confirmed `offer_approval_id`) that cannot be bypassed at the application layer.

13.4 `llm-gateway` Service

Routes all LLM calls across TalentAgent™:

Task	Model	Rationale
Candidate Intelligence Brief synthesis	GPT-4o	Long-context multi-source synthesis
JD generation, health scoring	GPT-4o	Narrative quality critical
NegotiationCoach™ scenario reasoning	GPT-4o	Game-theoretic reasoning quality
Outreach message generation	GPT-4o-mini	High volume; A/B monitored
Skill extraction, JD parsing, email parsing	Fine-tuned GPT-4o-mini	Low latency; high volume
Bias monitor, compliance reasoning	Claude 3.5 Sonnet	Constitutional AI for fairness-sensitive tasks
Screen response analysis	GPT-4o	Nuanced competency assessment

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14. Integration Registry — Full List

#	Integration	Module(s)	Auth	API Type	Cost Model	Build Phase
1	LinkedIn Talent Solutions API	2, 3	OAuth (partner)	REST	Partner pricing	Phase 1
2	GitHub REST API	2, 3, 5	OAuth 2.0 / API Key	REST	Free (rate limited)	Phase 1
3	Stack Exchange API	2	API Key	REST	Free	Phase 2
4	OpenAI API (GPT-4o, GPT-4o-mini, embeddings)	All	API Key	REST	Per token	Phase 1
5	Anthropic API (Claude 3.5 Sonnet)	10	API Key	REST	Per token	Phase 1
6	Pinecone	2, 3, 4	API Key	REST	Usage-based	Phase 1
7	Neo4j (Talent Intelligence Graph)	1, 2, 3, 7, 8	Bolt driver	Graph	Self-hosted / AuraDB	Phase 1
8	AWS SageMaker	2	AWS SDK	REST	Per inference hour	Phase 2
9	AWS Transcribe	5	AWS SDK	REST/WebSocket	Per audio second	Phase 1

#	Integration	Module(s)	Auth	API Type	Cost Model	Build Phase
10	AWS Cognito	All	AWS SDK	Managed	Per MAU	Phase 1
11	AWS SES	4, 7	AWS SDK	SMTP/REST	Per email	Phase 1
12	AWS EventBridge	2, 4, 6, 8	AWS SDK	Event	Per event	Phase 1
13	Apache Kafka (AWS MSK)	All	AWS MSK	Event stream	Per broker hour	Phase 1
14	Redis (ElastiCache)	All	AWS SDK	Cache	Per node hour	Phase 1
15	Greenhouse REST API + Webhooks	1, 2, 6	REST	REST + Webhooks	Free (requires Greenhouse license)	Phase 1
16	Lever REST API + Webhooks	1, 2, 6	REST	REST + Webhooks	Free (requires Lever license)	Phase 2
17	Workday Recruiting REST API	1, 7	OAuth 2.0	REST	Enterprise pricing	Phase 2
18	Workday HCM	7	OAuth 2.0	REST	Enterprise pricing	Phase 2
19	SAP SuccessFactors OData API	1, 8	OAuth 2.0 + middleware	OData/REST	Enterprise pricing	Phase 3
20	BambooHR API	10	API Key	REST	Per employee/month	Phase 3
21	Google Calendar API	4	OAuth 2.0	REST	Free	Phase 1
22	Microsoft Graph API (Outlook + Teams)	4, 6	OAuth 2.0 (Azure AD)	REST	Free	Phase 1
23	Calendly API	4	OAuth 2.0	REST	Per usage	Phase 2
24	Zoom API	6	OAuth 2.0	REST	Per meeting	Phase 1
25	daily.co API	6	API Key	REST	Per participant-minute	Phase 2
26	Glassdoor Partner API	3, 8	Partner agreement	REST	Partner pricing	Phase 1
27	Crunchbase Basic API	3	API Key	REST	Per request	Phase 1
28	Bing Web Search API	3	Azure API Key	REST	Per transaction	Phase 1
29	Levels.fyi	1, 7	Data license	REST	Annual license	Phase 2
30	Bureau of Labor Statistics API	1	Public	REST	Free	Phase 1
31	YouTube Data API v3	3	API Key	REST	Per quota unit	Phase 2

#	Integration	Module(s)	Auth	API Type	Cost Model	Build Phase
32	SpeakerDeck API	3	Public	REST	Free	Phase 2
33	LaunchDarkly	11	SDK	SDK	Per MAU	Phase 1
34	DocuSign API	7	OAuth 2.0	REST	Per envelope	Phase 2
35	Okta (SAML 2.0 / OIDC)	11	SAML / OIDC	Federation	Enterprise pricing	Phase 1
36	Azure AD (SAML 2.0 / OIDC)	11	SAML / OIDC	Federation	Per AAD license	Phase 1

15. AI Agent Registry — Full List

#	Agent Name	Module	LLM Model	Primary Tasks	Key Tools	Orchestration
1	`RequisitionIntelligenceAgent`	1	GPT-4o	JD generation, comp benchmarking, health scoring, approval workflow	jd_generator, comp_benchmark, health_scorer, approval_workflow, ats_sync	LangGraph conversational state machine
2	`RadarAgent`	2	GPT-4o-mini + GPT-4o	Multi-source discovery, fit scoring, passive signal detection, shortlist approval	linkedin_talent_api, github_search, ats_talent_pool, pinecone_match, neo4j_gnn, motivation_inference, passive_signal_detector	LangGraph parallel pipeline
3	`IntelligenceAgent`	3	GPT-4o	Master brief orchestration; synthesis of 4 sub-agents	brief_synthesiser, rag_pipeline, provenance_attacher, bias_monitor_gate	LangGraph pipeline
4	`LinkedInProfileSubAgent`	3	GPT-4o-mini	LinkedIn data extraction, career arc parsing, gap detection	linkedin_full_profile, career_arc_parser, gap_detector	Sub-agent of Module 3
5	`PublicWorkSubAgent`	3	GPT-4o	GitHub analysis, publication search, skills evidence mapping	github_full_profile, google_scholar, bing_search, youtube_speaker, skills_evidence_mapper	Sub-agent of Module 3
6	`MotivationInferenceSubAgent`	3	GPT-4o	Public writing analysis, motivation classification	content_sentiment_analyser, transition_pattern_analyser, motivation_classifier	Sub-agent of Module 3

#	Agent Name	Module	LLM Model	Primary Tasks	Key Tools	Orchestration
7	`ConversationAngleSubAgent`	3	GPT-4o	Interview angle generation, verification question creation	profile_role_crossref, interview_angle_generator	Sub-agent of Module 3
8	`OutreachAgent`	4	GPT-4o-mini	Personalised message generation, send queue management, response rate learning	brief_retriever, signal_extractor, message_generator, ses_send, thompson_sampling	LangGraph approval gate + A/B loop
9	`SchedulerAgent`	4	GPT-4o-mini	Calendar negotiation, interview booking, rescheduling, no-show detection	email_availability_parser, google_calendar, outlook_calendar, calendly_fallback, no_show_alerter	LangGraph event-driven state machine
10	`ScreenAgent`	5	GPT-4o	Question generation, video response analysis, portfolio analysis, capability profile	screen_question_generator, aws_transcribe, response_analyser, github_portfolio_analyser, capability_profile_generator	LangGraph sequential pipeline
11	`InterviewOrchestrationAgent`	6	GPT-4o	Pre-interview pack, video hosting, debrief collection, outcome communication	pre_interview_pack, zoom_api, teams_api, debrief_collector, outcome_communicator	LangGraph event-driven
12	`OfferAgent`	7	GPT-4o	Offer package generation, scenario modelling, offer letter, approval workflow	workday_comp_band, neo4j_benchmark, scenario_modeller, offer_approval, offer_letter_generator	LangGraph multi-gate state machine
13	`AnalyticsAgent`	8	GPT-4o-mini	Pipeline metrics, diversity analytics, CHRO report generation	pipeline_aggregator, diversity_calculator, platform_benchmarks, chro_report_generator, hris_push	Scheduled EventBridge + real-time WebSocket

#	Agent Name	Module	LLM Model	Primary Tasks	Key Tools	Orchestration
14	`NegotiationCoachRecruiterAgent`	9	GPT-4o	Real-time negotiation advisory, comp band guardrails, scenario reasoning	websocket_recruiter_stream, offer_benchmark_display, comp_band_guardrail, scenario_advisor_recruiter	Persistent WebSocket session
15	`BiasMonitorAgent`	10	Claude 3.5 Sonnet	Disparate impact testing, SHAP explainability, corrective re-ranking	disparate_impact_tester, proxy_demographic_classifier, shap_explainer, corrective_ranker, bias_report_generator	Sidecar container; intercepts at pipeline checkpoints
16	`ComplianceAgent`	10	Claude 3.5 Sonnet	DSAR processing, compliance reporting, EU AI Act monitoring	dsar_data_exporter, pinecone_vector_deletion, neo4j_pii_clearer, compliance_report_generator, eu_ai_act_disclosure	Event-driven Lambda + scheduled EventBridge

16. Build Complexity & Cost Matrix

16.1 Development Complexity by Module

Module	Frontend Complexity	Backend Complexity	AI/Agent Complexity	Overall	Recommended Phase
1 — Requisition Intelligence	Medium	High	High	**High**	Phase 1 (Month 1–3)
2 — Candidate Sourcing	Medium	Very High	Very High	**Very High**	Phase 1 (Month 2–4)
3 — Candidate Intelligence Brief™	High	High	Very High	**Very High**	Phase 1 (Month 3–5)
4 — Outreach & Scheduling	Medium	High	High	**High**	Phase 1 (Month 3–5)
5 — Screen & Assessment	High	High	Very High	**Very High**	Phase 1 (Month 4–6)
6 — Interview Orchestration	Medium	High	Medium	**High**	Phase 1 (Month 5–6)
7 — Offer Management	High	High	High	**High**	Phase 2 (Month 6–8)
8 — Analytics & Reporting	High	High	Medium	**High**	Phase 2 (Month 6–8)
9 — NegotiationCoach™	High	Very High	High	**Very High**	Phase 2 (Month 8–10)
10 — Compliance & Bias	Low	Very High	High	**Very High**	Phase 1 foundational + Phase 2 depth
11 — Admin Console	Medium	High	Low	**Medium–High**	Phase 1 (Month 2–4)

16.2 Recurring Operational Cost Drivers (Monthly, at Scale)

Cost Category	Driver	Estimated Cost (at 50 enterprise seats)	Notes
LLM API — GPT-4o	CIB generation, NegotiationCoach™, offer generation, screen analysis	$15,000–$30,000/month	CIB = ~80k tokens per candidate; NegotiationCoach live = ~30k tokens per session
LLM API — GPT-4o-mini	Outreach generation, JD parsing, email parsing, skill extraction	$2,000–$5,000/month	High volume; low cost per call
LLM API — Claude 3.5 Sonnet	Bias monitor, compliance reports, DSAR processing	$1,000–$3,000/month	Lower volume; premium justified for fairness-critical tasks
Pinecone	Candidate embedding storage and semantic search	$1,000–$3,000/month	Per-tenant namespace; scales with candidate volume
Neo4j AuraDB	Talent Intelligence Graph	$2,000–$5,000/month	Professional tier
AWS infrastructure (EKS, RDS, ElastiCache, MSK, SageMaker)	All services	$5,000–$12,000/month	GPU nodes for SageMaker inference; Phase 2+
LinkedIn Talent Solutions	Candidate profile data	$5,000–$20,000/month	Partner pricing; heavily usage-dependent — single largest API cost
Glassdoor Partner API	Company + interview data	$1,000–$3,000/month	Partner pricing
AWS Transcribe	Screen response transcription	$500–$2,000/month	Per audio minute; scales with screening volume
Zoom API	Interview hosting	$500–$1,500/month	Per meeting; scales with interview volume

Total estimated operational cost at 50 enterprise seats: ~$33,000–$84,000/month (Before enterprise subscription revenue; LinkedIn Talent Solutions is the most volatile cost — model carefully during commercial planning)

16.3 One-Time Build Cost Drivers

Item	Estimated Engineering Cost	Notes
Talent Intelligence Graph initial build (Neo4j + ESCO/O*NET)	$30,000–$50,000	Shared with CareerAgent™ build
GraphSAGE GNN model training	$20,000–$40,000	Phase 2; requires curated training data
Fine-tuning GPT-4o-mini for JD parsing + skill extraction	$10,000–$20,000	Training data curation + runs
LinkedIn Talent Solutions partnership	$0–$10,000 (legal + BD)	Required for all LinkedIn profile data
Greenhouse API certification	$0–$5,000	Requires Greenhouse partner review
Glassdoor data partnership	$0–$5,000	Glassdoor partner agreement
Levels.fyi data license	$10,000–$30,000/year	Annual fee
Bias testing framework development	$20,000–$40,000	Proxy demographic classifier; SHAP pipeline; corrective ranker
EU AI Act compliance documentation (Month 6–24)	$30,000–$60,000	Technical documentation, Notified Body engagement

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17. Development Roles Required

Role	Modules	FTE Estimate (Phase 1)	Notes
AI/ML Engineer	1–7 agent modules	3 FTE	LangGraph, LLM integration, RAG pipeline, NegotiationCoach™
Backend Engineer (Python)	agent-orchestrator, intelligence services, bias monitor, compliance	2 FTE	FastAPI, LangGraph, Kafka, bias architecture
Backend Engineer (Node.js)	recruiter-service, ATS connectors, integration-hub, offer-service	2 FTE	Express, PostgreSQL, ATS integrations
Frontend Engineer (React 18)	All recruiter dashboard surfaces	2 FTE	React 18, Zustand, Recharts, D3.js, TipTap
ML Engineer (Graph/GNN)	Talent Intelligence Graph, GraphSAGE, SHAP	1 FTE (Phase 2)	Neo4j, GraphSAGE, SageMaker, SHAP
Data Engineer	Kafka → S3/Athena pipeline, Pinecone embeddings, Neo4j data import	1 FTE	ETL, embedding pipeline, data warehouse
DevOps / Platform Engineer	EKS, CI/CD, observability, multi-tenant infra	1–2 FTE	Kubernetes, Helm, Terraform, Prometheus/Grafana
Security Engineer	Auth (Cognito + SSO), KMS, GDPR, EU AI Act	1 FTE	SAML/OIDC, encryption architecture, compliance
Product Designer	All dashboard surfaces + Admin Console	1 FTE	Enterprise design system, accessibility
BD / Partnership Lead	LinkedIn, Glassdoor, Greenhouse, Levels.fyi partnerships	1 FTE (non-engineering)	Partnership agreements required before API access

Minimum Phase 1 team: 12–14 engineers + 1 designer + 1 BD lead

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18. Phase-by-Phase Delivery Plan

Phase 1 — Core Platform (Months 1–6)

Goal: Operational TalentAgent™ with requisition intelligence, candidate sourcing, Candidate Intelligence Brief™, outreach, scheduling, screening, and interview orchestration. All with Human Handoff Protocol and basic compliance infrastructure.

Sprint	Deliverable	Module
Sprint 1–2	Infrastructure: EKS cluster, RDS, Redis, Kafka (MSK), Cognito (SSO), CI/CD, LaunchDarkly	Cross-cutting
Sprint 3–4	`recruiter-service`, requisition schema, RBAC, Admin Console (basic)	Module 1, 11
Sprint 5–6	RequisitionIntelligenceAgent: JD generation, comp benchmarking, health scoring	Module 1
Sprint 7–8	Greenhouse connector, ATS sync framework, `integration-hub`	Module 1, 11
Sprint 9–10	RadarAgent (v1 — semantic matching only), LinkedIn integration, candidate shortlist UI	Module 2
Sprint 11–12	IntelligenceAgent pipeline: all 4 sub-agents, brief synthesis, provenance UI	Module 3
Sprint 13–14	BiasMonitorAgent (sidecar), disparate impact test, SHAP explainability	Module 10
Sprint 15–16	OutreachAgent: message generation, approval gate, send queue	Module 4
Sprint 17–18	SchedulerAgent: Google Calendar + Outlook integration, confirmation emails	Module 4
Sprint 19–20	ScreenAgent: question generation, video infrastructure (AWS Transcribe), response analysis	Module 5
Sprint 21–22	InterviewOrchestrationAgent: pre-interview pack, Zoom/Teams integration, debrief flow	Module 6
Sprint 23–24	Integration testing, performance optimisation, security review, soft launch to beta customers	All

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Phase 2 — Intelligence Depth & Offer Management (Months 7–12)

Goal: GNN match scoring, offer management, NegotiationCoach™, analytics reporting, Lever integration.

Sprint	Deliverable	Module
Sprint 25–26	GNN match scoring (GraphSAGE, SageMaker), RadarAgent upgrade	Module 2
Sprint 27–28	OfferAgent: offer package generation, scenario modelling, approval workflow	Module 7
Sprint 29–30	Database-level offer constraint, offer letter generation, DocuSign integration	Module 7
Sprint 31–32	NegotiationCoachRecruiterAgent: live session, WebSocket channels, mutual acknowledgment gate	Module 9
Sprint 33–34	AnalyticsAgent: pipeline dashboard, diversity analytics, CHRO report PDF	Module 8
Sprint 35–36	Lever connector, Workday Recruiting integration	Module 1, 2
Sprint 37–38	Workday HCM comp band integration, NegotiationCoach™ comp guardrails	Module 7, 9
Sprint 39–40	Passive candidate detection upgrade, motivation inference model tuning	Module 2

Phase 3 — Enterprise Scale, Compliance & EU Expansion (Months 13–24)

Goal: EU AI Act compliance, GDPR data residency, enterprise hardening, additional ATS connectors, HOGH Public API.

Sprint	Deliverable
Sprint 41–44	GDPR data residency (eu-west-1 deployment), DSAR automation, EU AI Act technical documentation (Month 6 obligation)
Sprint 45–48	EU AI Act transparency tooling: candidate disclosure portal, opt-out mechanism, data provenance reports
Sprint 49–52	SAP SuccessFactors OData connector, BambooHR integration
Sprint 53–56	GPT-4o-mini fine-tuning: JD parsing + skill extraction (Phase 3 cost optimisation)
Sprint 57–60	HOGH Talent Intelligence API (public API layer): `/v1/match`, `/v1/skill-graph/query`, `/v1/benchmarks/compensation`
Sprint 61–68	EU AI Act formal conformity assessment, Notified Body engagement (Month 24 target)

Summary Timeline

Phase	Duration	Primary Output	Key Risk
Phase 1	Months 1–6	Core TalentAgent™ in production; beta enterprise customers	LinkedIn partnership approval time (allow 3–6 months for LinkedIn Talent Solutions agreement)
Phase 2	Months 7–12	Full pipeline including offer management and NegotiationCoach™	GNN training data quality; Workday integration complexity
Phase 3	Months 13–24	EU compliance, APAC prep, public API, fine-tuned models	EU AI Act Notified Body timeline; SAP SuccessFactors integration complexity

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